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4.0 PHYLUM: CHORDATA

Chordates are one of the most familiar groups of animals to mankind. Phylum **Chordata** includes all those animals that possess the notochord (a primary endoskeletal element, their **chief distinctive** character) at some stage in their life time. It is one of the major animal phyla in terms of the number of species. Chordates are believed to have descended from **echinoderm larvae-like ancestors** during the **Precambrian** period. Chordates include **protochordates** (tunicates, lancelets) and **vertebrates** (cyclostomes, fishes, amphibians, reptiles, aves and mammals). Most of them are aquatic and some are amphibious or terrestrial animals. Though they exhibit variation in the size and shape of their bodies, they display similarity in their basic body plan. The first and the second largest of the living animals, namely blue whale (***Balaenoptera musculus*** - a marine mammal) and whale shark (***Rhinodon typus***) are members of the phylum Chordata.

I. The four principal chordate characters

1. Notochord

It is a flexible rod like structure situated along the mid dorsal line between the gut and the nerve cord. It is the first part of the endoskeleton to appear in the embryo. It is derived from the embryonic **chorda mesoderm**. It is made up of a core

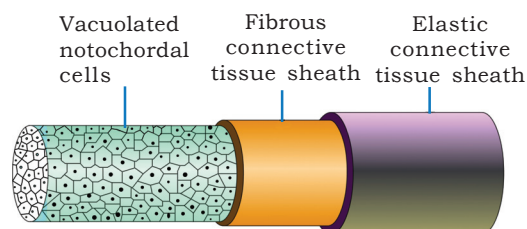


Fig. 4.1 Longitudinal view of notochord

of 'vacuolated cells' surrounded by an inner thick **fibrous** and an outer thin **elastic** connective tissue sheaths. It is persistent throughout their life in the **lancelets**, **cyclostomes** and fishes of the group chondrichthyes (NCERT). It is present in the tail of the tadpole larva of an **ascidian** which is lost in its adult stage due to **retrogressive metamorphosis** (see Glossary). It is present in the embryonic stages, but is replaced partly or wholly by the vertebral column in the adults of higher chordates. Remnants of notochord occur as **nuclei pulposi** in the **intervertebral discs** of mammals.

2. Dorsal tubular nerve cord

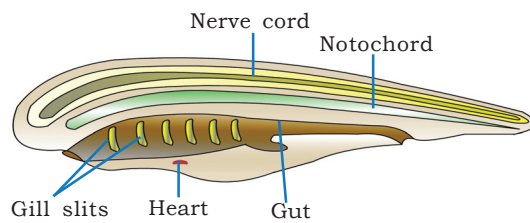


Fig. 4.3 Typical chordate

A single, hollow tubular and fluid filled nerve cord is situated above the notochord and below the dorsal body wall. It is **non-ganglionated**, unlike that of the non-chordates. It is produced in the embryonic stage by the 'sinking in' of the median dorsal strip of **ectoderm** above the **notochord**. In the higher chordates, it gets enlarged to form a distinct **brain** at the anterior end and the rest of it becomes the **spinal cord**.

3. Pharyngeal slits or clefts

Pharyngeal slits are a series of lateral perforations in the wall of the pharynx through which water flows out from the pharyngeal cavity. They are **ecto-endodermal** in origin. They are persistent throughout the life in the protochordates, fishes and some amphibians. The walls of the pharyngeal slits develop vascular lamellae to become 'gills' (branchiae) and are helpful in the exchange of respiratory gases. They are present in the larval stages, but are absent in the adults of many amphibians. Among the amniotes, non-functional pharyngeal pouches appear in the early embryonic life and disappear later. The occurrence of such structures in the early embryos of amniotes provides a clue to their 'aquatic ancestry'.

4. Post-anal tail

Chordates have a tail extending posterior to the anus. It is lost in many species during the late embryonic development. It contains skeletal elements and muscles. However, coelom and visceral organs are absent in it. It provides the propelling force in the locomotion of many aquatic species and acts as a 'balancing organ' in some terrestrial forms such as Kangaroos.

Chordates share **deuterostomeate condition**, **radial** and **indeterminate** cleavage and **enterocoelom** with the echinoderms.

II. Other chordate characters

Bilateral symmetry, triploblastic condition, organ-system level of organisation, metamerism, cephalisation, 'enterocoelic' coelom (except in the higher chordates in which it is secondarily schizocoelic), 'complete' digestive tract, closed circulatory system, ventral 'myogenic' heart (except in the lancelets), phosphocreatine (muscle phosphagen), etc., constitute the other important features of the chordates.

Phylum Chordata is divided into three subphyla: **Urochordata** or **Tunicata**, **Cephalochordata** and **Vertebrata**. Urochordates and Cephalochordates are together referred to as '**protochordates**'.

III. Comparison of Chordates and Non-chordates

S.No.	Chordates	Non-chordates
1.	Notochord is present.	Notochord is absent.
2.	Central nervous system is dorsal, hollow, single and non-ganglionated.	Nerve cord is ventral, solid, double and ganglionated.
3.	Pharynx is perforated by gill slits.	Pharyngeal gill slits are absent.
4.	Heart is ventral.	Heart is dorsal (if present).
5.	A post-anal tail is present.	Post-anal tail is absent.

4.1 SUBPHYLUM-UROCHORDATA OR TUNICATA

(**GR.OURA-** A TAIL; **L. CHORDA-** CORD)

All urochordates are marine and occur from the surface water to greater depths. They are either sessile (ascidians) or pelagic (*Salpa*, *Doliolum*) and solitary (*Ascidia*) or colonial (*Pyrosoma*). Body is un-segmented and covered by a **test** or **tunic** composed of **cellulose**¹ which is uncommon in animals. Coelom is absent. However, an ectoderm-lined **atrial cavity**, surrounds the pharynx, into which gill slits, anus and genital ducts open. Ventral side of the pharynx possesses an **endostyle**, which is believed to be the forerunner of the thyroid gland of a vertebrate (see Glossary). Atrium leads to the exterior by a dorsal or posterior atrial aperture. Digestive tract is complete. Two to numerous gill slits are found in the pharyngeal wall. Circulatory system is of **open type**. There is a simple, tubular, **ventral heart** which alternately reverses the direction of the flow of blood. Nervous system is represented in the adult by a single dorsal 'ganglion'. They are **bisexual** or **hermaphrodites**. Development generally includes a free-swimming **tadpole larva** with a tail, a dorsal hollow nerve cord, and a notochord confined to the tail, hence the name **urochordata**.

e.g. *Ascidia*, *Salpa*, *Doliolum*, *Pyrosoma* and *Oikopleura*.

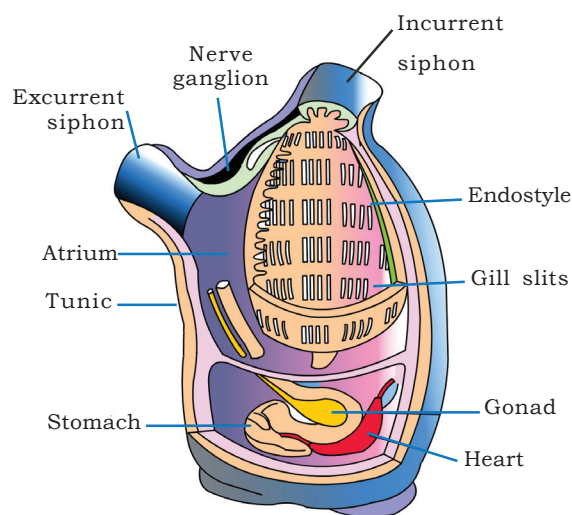


Fig. 4.4 *Ascidia*– Internal organisation

Is it necessary to know the life history of an ascidian to consider it a chordate?

¹ Ref. Integrated Principles of Zoology 12th Edition by Hickman and Biology 9th Edition by Raven and Johnson.